

Last time

cts. f on $[a, b] \Rightarrow \forall \varepsilon > 0, \exists \delta > 0$ s.t. whenever $|x - y| < \delta$, $|f(x) - f(y)| < \varepsilon$

f unif. cts. on $[a, b]$

Pf. Let $\varepsilon > 0$

Define $A_\varepsilon := \{x \in [a, b] : \exists \delta_1 > 0 \text{ whenever } |x' - x''| < \delta_1, \text{ we have } |f(x') - f(x'')| < \varepsilon\}$

$A_\varepsilon \neq \emptyset$ b/c. $a \in A_\varepsilon$

By construction, A_ε bounded above by b .

$\Rightarrow \alpha := \text{LUB}(A_\varepsilon)$ exists

If $\alpha < b$

Since f cts. at α , for $\frac{\varepsilon}{2}$, $\exists \delta_2$ s.t. whenever $|x - \alpha| < \delta_2$, $|f(x) - f(\alpha)| < \frac{\varepsilon}{2}$.

$$\delta = \frac{1}{2} \min\{\delta_1, \delta_2\}$$

Claim for our $\varepsilon > 0$. whenever $|x' - x''| < \delta$, $x', x'' \in [a, \alpha + \delta_2]$

We have $|f(x') - f(x'')| < \varepsilon$.

• If $x', x'' \in [a, \alpha)$, $|x' - x''| < \delta < \delta_1$

then by construction of A_ε , $|f(x') - f(x'')| < \varepsilon$.

• If $x', x'' \in (\alpha - \delta_2, \alpha + \delta_2)$ and $|x' - x''| < \delta$

then $|f(x') - f(x'')|$

$$\leq |f(x') - f(\alpha)| + |f(\alpha) - f(x'')|$$

$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$$

$$= \varepsilon$$

It is impossible to have $x' \in [a, a - \delta_2)$, $x'' \in (a, a + \delta_2)$

$$\Rightarrow |x' - x''| \geq \delta_2$$

but we're assuming $|x' - x''| < \delta < \delta_2$

$$\delta_2 > |x' - x''| \geq \delta_2$$

$$\delta_2 > \delta_2, \text{ contradiction.} \Rightarrow a = b$$

So it remains to check that $b \in A_\varepsilon$

This is similar to the pf of EVT and also the argument that we did.



Ex An example of a function that is uniform cts. but w/ unbounded derivative.

$$f(x) = x^{1/3} \text{ on } [-1, 1]$$

$$f'(x) \rightarrow \infty \text{ as } x \rightarrow 0$$

but since f is cts. on $[-1, 1]$, by prev. thm. It is unif. cts.

XXII. Infinite Sequences

Def. An infinite sequence is a function $f: \mathbb{Z}_{>0} \rightarrow \mathbb{R}$
 $n \mapsto f(n) := a_n$

Instead of writing it as a function, we write $(a_n)_{n \in \mathbb{Z}_{>0}}$

Def. A seq (a_n) converges to $l \in \mathbb{R}$ ($\lim_{n \rightarrow \infty} a_n = l$) if $\forall \varepsilon > 0, \exists N$ s.t. $\forall n > N$
 $|a_n - l| < \varepsilon$

Ex ① $a_n = \frac{1}{n}$ ($\lim_{n \rightarrow \infty} a_n = 0$)

To prove this, let $\varepsilon > 0$

let N be an integer $> \frac{1}{\varepsilon}$

then for any $n > N$, $|a_n - 0| = |a_n| = \frac{1}{n} < \varepsilon$



② $a_n = n$. This does not converge to any l .

Suppose for a contradiction $\lim_{n \rightarrow \infty} a_n = l$ for some l

Let $\varepsilon = 1$. Suppose some N exists, some $n > N$ s.t. $|a_n - l| > 1$
 $\parallel$
 $|n - l|$

So any $n > l+1$ doesn't satisfy this, so let $n = \max\{l+1, N\} + 1$

Remark If (a_n) doesn't converge to anything, then it diverges.